

Balaji Temple Scholarship Application System

System Flow & Architecture Layout

1. OVERVIEW

The Balaji Temple Scholarship Application System is designed as a secure, multi-step process that allows students to apply for scholarships while maintaining data integrity through two-factor authentication and progressive form completion.

2. STUDENT APPLICATION WORKFLOW

Phase 1: Initial Registration

Step 1: Basic Application Form

- Student visits the scholarship portal
- Completes basic application with:
 - Personal Information (Name, DOB, Gender, etc.)
 - Contact Details (Email, Phone, Address)
 - High School/Academic Information
 - Parent/Guardian Information
- Submits the initial application

Step 2: Email Verification & Authentication

- System generates unique Application ID
- System generates One-Time Password (OTP) for security
- Email sent to registered email address containing:
 - Application ID
 - OTP (Time-limited)
 - Login instructions
- Student receives confirmation email

Phase 2: Secure Login

Step 3: Two-Factor Authentication Login

- Student navigates to Student Login page
- Required credentials:
 - Application ID
 - Registered Email Address
 - OTP from email
- System validates all three credentials
- If validation fails: Error message displayed with retry option
- If validation succeeds: Student redirected to Dashboard

Phase 3: Complete Additional Information

Step 4: Student Dashboard

The dashboard displays all required sections with visual status indicators:

-  **Completed sections** - Grayed out (view-only)
-  **Pending sections** - Active and editable

Required Sections:

1. Financial Information

- Family income details
- Assets information
- Financial aid received from other sources
- Household expenses

2. Education Information

- Academic records
- GPA/Test scores
- College admission details
- Intended major/program
- Academic achievements

3. Document Upload

- Academic transcripts
- Financial documents (tax returns, W-2s)
- Admission letters
- Photo ID
- Other supporting documents
- **Documents saved to SharePoint** in folder named with Application ID

4. Questionnaires

- Essay questions
- Community service information
- Temple involvement
- Personal statements
- Future goals

5. Additional Financial Information

- Detailed financial need explanation
- Special circumstances
- Other scholarships applied for

Step 5: Progressive Completion

- Student can complete sections in any order
- Each completed section:
 - Automatically saved to database
 - Grayed out on dashboard
 - Status updated to "Completed"
- Student can save progress and logout at any time

Phase 4: Re-authentication (If Needed)

Step 6: Returning Students

Scenario A: OTP Still Valid

- Student logs in with: Application ID + Email + Original OTP
- Redirected to dashboard with completed sections intact

Scenario B: OTP Expired

- Student enters: Application ID + Email on reauthentication page
- System generates and sends new OTP
- Student receives new OTP via email
- Student logs in with new OTP
- Redirected to dashboard showing previous progress

Phase 5: Final Submission

Step 7: Application Completion

- Dashboard checks if all sections are completed
 - If incomplete: Student cannot submit final application
 - If complete: "Submit Final Application" button enabled
 - Student reviews and submits
 - Confirmation message displayed
 - Confirmation email sent
 - Application status updated to "Submitted"
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3. SYSTEM PROCESSING LAYER

Authentication System

- **Application ID Generation:** Unique identifier for each application
- **OTP Generation:** Time-limited (e.g., 30 minutes), single-use passwords
- **Two-Factor Authentication:** Validates Application ID + Email + OTP
- **Session Management:** Tracks active sessions and expiration

Data Management

- **Database Storage:**
 - All application data stored in secure database
 - Real-time status tracking for each section
 - Timestamp for each submission
 - Application progress tracking
- **SharePoint Integration:**
 - Document storage organized by Application ID
 - Folder structure: `/ApplicationID/[uploaded files]`
 - Secure access controls
 - Version tracking for documents

Status Tracking

- **Section Status:** Pending, In Progress, Completed
 - **Application Status:** Draft, In Progress, Submitted, Under Review, Approved/Rejected
 - **Automated Updates:** Status changes trigger email notifications
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4. ADMIN PORTAL FUNCTIONALITY

Admin Dashboard Features

Application Grid View Displays comprehensive data in tabular format:

- Application ID
- Student Name
- Email Address
- Phone Number
- Date of Initial Application
- Section Completion Status (5 sections)
- Overall Completion Percentage
- Submission Date
- Application Status
- Review Status
- Actions (View/Edit/Update)

Admin Actions

1. **View Application Details**

- Full student profile
- All submitted information across sections
- Application timeline
- Communication history

2. **Access Documents**

- Direct links to SharePoint folders
- View uploaded documents
- Download individual or bulk files
- Document verification status

3. **Update Application Status**

- Change review status
- Add reviewer notes
- Approve/Reject applications
- Request additional information

4. **Generate Reports**

- Application statistics
- Completion rates
- Financial need analysis
- Export to Excel/PDF
- Custom filtered reports

5. **Communication**

- Send emails to students
- Bulk communication options
- Template-based messages

5. DATA FLOW ARCHITECTURE

Student → System

1. Student submits data via web forms
2. Data validated on client-side (JavaScript)
3. Data validated on server-side
4. Data stored in database with encryption
5. Documents uploaded to SharePoint
6. Email notifications triggered
7. Status updates reflected in real-time

System → Admin

1. Admin logs into portal
 2. Database queried for application data
 3. Grid populated with current information
 4. SharePoint API called for document access
 5. Admin views/edits as needed
 6. Changes saved back to database
 7. Notifications sent to students if needed
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6. SECURITY FEATURES

Authentication

- Two-factor authentication (Email + OTP)
- Time-limited OTP (prevents reuse)
- Session timeout after inactivity
- Secure password storage (if implemented)

Data Protection

- SSL/TLS encryption for all data transmission
- Database encryption at rest
- Role-based access control (Student vs. Admin)
- Audit logging for all data changes
- SharePoint security permissions

Privacy

- PII (Personally Identifiable Information) protection
 - Access logs maintained
 - Data retention policies
 - GDPR/Privacy compliance considerations
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7. TECHNICAL REQUIREMENTS

Front-End

- Responsive web design (mobile-friendly)
- Form validation
- Progress indicators
- File upload with size/type restrictions
- Real-time status updates

Back-End

- Application ID and OTP generation
- Email service integration (SMTP)
- Database management (SQL Server/MySQL)
- SharePoint API integration
- Session management
- RESTful API for data operations

Integration Points

- Email Server (for OTP and notifications)
 - SharePoint Online (for document storage)
 - Database (for application data)
 - Payment Gateway (if scholarship payments are processed)
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8. USER EXPERIENCE ENHANCEMENTS

Student Experience

- Clear progress indicators
- Save and continue later functionality
- Email reminders for incomplete applications
- Help text and tooltips
- Document upload preview
- Mobile-responsive design
- Auto-save draft functionality

Admin Experience

- Intuitive dashboard
 - Advanced search and filters
 - Bulk actions capability
 - Export functionality
 - Quick access to documents
 - Application comparison tools
 - Automated notifications
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9. NOTIFICATION SYSTEM

Student Notifications

1. Initial application confirmation
2. OTP for login
3. Section completion confirmations
4. Final submission confirmation
5. Application status updates
6. Request for additional information
7. Scholarship decision notification

Admin Notifications

1. New application received
2. Application submitted (ready for review)
3. Incomplete applications reminder
4. Review deadline reminders

10. TIMELINE & MILESTONES

Application Timeline

- **Basic Application:** 10-15 minutes
- **Additional Sections:** 1-2 hours (can be spread over multiple sessions)
- **Document Upload:** 15-30 minutes
- **Review & Submit:** 10 minutes

System Timeline

- **OTP Validity:** 30 minutes (configurable)
 - **Session Timeout:** 30 minutes of inactivity
 - **Application Deadline:** May 27, 2025 (as per document)
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11. ERROR HANDLING

Common Scenarios

1. **Invalid OTP:** Clear error message, option to resend
 2. **Expired OTP:** Automatic reauthentication flow
 3. **Incomplete Upload:** Retry mechanism with error details
 4. **Session Timeout:** Save progress, require re-login
 5. **Network Issues:** Auto-save drafts, resume capability
 6. **Duplicate Application:** Check and prevent duplicate submissions
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12. FUTURE ENHANCEMENTS (Optional)

- SMS OTP option in addition to email
 - Social media integration for easier login
 - Video essay upload capability
 - Interview scheduling integration
 - Peer recommendation system
 - Mobile application
 - Real-time chat support
 - AI-powered application review assistance
 - Automated financial need calculation
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13. SUCCESS METRICS

Key Performance Indicators

- Application completion rate
 - Average time to complete application
 - Document upload success rate
 - OTP validation success rate
 - User satisfaction score
 - Admin processing time
 - Number of applications received vs. scholarships awarded
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APPENDIX

Application Sections Detailed Breakdown

Basic Application Fields:

- First Name, Middle Name, Last Name
- Date of Birth
- Gender
- Email Address (primary communication)
- Phone Number
- Current Address
- Permanent Address
- High School Name and Details
- Expected Graduation Date
- Parent/Guardian Information

Financial Information Fields:

- Annual Family Income
- Number of Dependents
- Assets Owned
- Other Financial Aid Received
- Expected Family Contribution
- Special Financial Circumstances

Education Information Fields:

- Academic GPA
- SAT/ACT Scores
- Class Rank
- College Acceptance Status
- Intended University
- Major/Program
- Academic Honors and Awards
- Extracurricular Activities

Required Documents:

- High School Transcript
- SAT/ACT Score Report
- College Acceptance Letter
- Parent/Guardian Tax Returns (Previous Year)
- W-2 Forms
- Photo Identification
- Any Additional Supporting Documents

Questionnaire Topics:

- Why do you deserve this scholarship?
 - How has Balaji Temple impacted your life?
 - Community service involvement
 - Future career goals
 - How will this scholarship help you achieve your goals?
 - Describe a challenge you've overcome
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CONCLUSION

This comprehensive system design ensures a secure, user-friendly, and efficient scholarship application process while maintaining data integrity and providing administrators with powerful tools to manage and review applications effectively.